8 B. Adaptations of Xerophytes - *Opuntia*

Opuntia is a succulent or drought resisting xerophyte, which grows wild in arid areas.

Diagnostic Features

- The stem is flattened, green, thick and fleshly called phylloclade
- Mucilage is present which helps to retain the water.
- Leaves are modified into spines

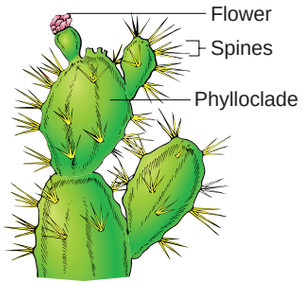


Figure 8a: Succulent xerophyte - *Opuntia*

8 C. Adaptation of Halophytes – Pneumatophores of *Avicennia*

Avicennia is a plant which grows and survives in saline environment like salty lakes and sea shores (mangrove vegetation).

Diagnostic Features

- A special kind of negatively geotropic root called pneumatophores (respiratory roots) are present.
- The leaves excrete salts through the salt glands.

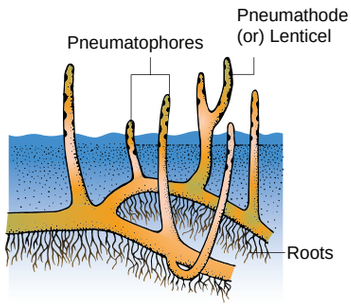


Figure 8c: Pneumatophores of *Avicennia*

8 D. Adaptation of Epiphytes – *Vanda*

Vanda is an epiphytic plant that grows perched on other plants (supporting plants). They use supporting plants only as shelter and not for water or food supply.

Diagnostic Features

- Root system is extensively developed. These roots are of two types (1) clinging roots and (2) aerial roots.
- The clinging roots fix epiphytes firmly on the surface of the supporting plant.
- Aerial roots are green coloured, hang downwardly and absorbs moisture from the atomosphere with the help of spongy tissue called velamen.

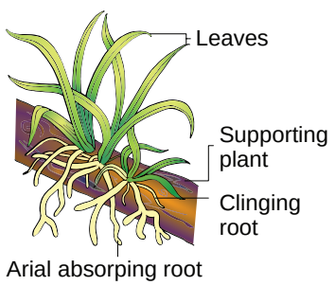


Figure 8d: Epiphytic roots of *Vanda*

III - Models / Photographs / Pictures

Exercise 9: Types of ovules

Aim: To recognize different types of ovules in flowering plants

Principle: To identify ovules based on the orientation, form and position of the micropyle with respect to funicle and chalaza

Requirements: Models / Photographs/ Pictures of different types of ovules.

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